

Product measure and independence

Topic 4

Outline

- 1 Produce measure
- 2 Fubini-Tonellis's theorem
- 3 Independence

Through this topic, you should...

After finishing this topic, you should know:

- Definitions produce measure.
- Funini-Tonellis's theorem.
- Independence in the language of probability.
- Be familiar to multivariate normal random variables.

Joint information and marginal laws

Topic 3 used independent samples. We now ask how to construct their joint probability space.

Let $U \sim \text{Unif}(0, 1)$. Compare the joint laws of (U, U) and $(U, 1 - U)$.

Example. Compare the marginal laws and the probabilities of $D = \{(x, y) : x = y\}$.

- Why we need to introduce “independent” random variables?
- The process of collecting samples should be representative fairness of the samples. We can fairly draw samples from the sample space.
- In many probability theorems, law of large numbers, central limit theorem... (we will see them later), we always assume random variables are independently and identically distributed.
- How we mathematically say independence?
- Before we introduce and study independent random variables, we need some further background in measure theory. More specifically, given two probability spaces $(\Omega_1, \mathcal{F}_1, \mathbb{P}_1)$ and $(\Omega_2, \mathcal{F}_2, \mathbb{P}_2)$, we will construct a product probability measure $\mathbb{P}_1 \times \mathbb{P}_2$ on the product space $\Omega_1 \times \Omega_2$. We will then extend this construction to an infinite product space $\Omega_1 \times \Omega_2 \times \cdots$, which will allow us to construct an infinite sequence of independent random variables.

- **Definition(Product σ -algebra).**
- The product of $(A_1, \mathcal{F}_1)$ and $(A_2, \mathcal{F}_2)$ is the measurable space
- Remark that Cartesian product of two σ -algebras is not a σ -algebra in general.

- **Theorem(Product measure).**
- We omit the proof, the proof needs Dykin's $\pi - \lambda$ theorem.
- **Definition(A Set and its slices).** Let $(\Omega_1, \mathcal{F}_1)$, $(\Omega_2, \mathcal{F}_2)$ be two finite (or σ -finite) measurable spaces, and the set $A \in \mathcal{F}_1 \otimes \mathcal{F}_2$, the product σ -algebra on $\Omega_1 \times \Omega_2$. Let

denote the ω_1 -slice and ω_2 -slice of A , respectively.

- Similarly, if $f : \Omega_1 \times \Omega_2 \rightarrow \mathbb{R}$ and $\omega_i \in \Omega_i$, then we can define

Introduction to Fubini's theorem

- Also, we can conclude that $P_1(A_{\omega_2})$ and $P_2(A_{\omega_1})$ are $\mathcal{F}_2$ -measurable and $\mathcal{F}_1$ -measurable, respectively, and
- Imaging we are calculating the area of a rectangle
- **Example.** Can you come up with an example such that a **non-measurable set** with **measurable slices**?
- From the example, we have $A \subset \Omega_1 \times \Omega_2$ is **not measurable** with respect to the product σ -algebra $\mathcal{F}_1 \otimes \mathcal{F}_1$ but its slices A_{ω_1} and A_{ω_2} are **measurable** for all ω_1 and ω_2 .

Fubini-Tonellis's theorem

- By definition a function f is called Lebesgue integrable if f is measurable and $\int_E |f(x)|\mu(dx) < \infty$
- Measurable and integrable: If function f is bounded on set E of finite measure, then f is Lebesgue integrable over E if and only if f is measurable. For example,
- We remark that the relation between measurable and integrable are not easy in general σ -finite measure to say without future measure theory background, but we are “safe” in probability space.
- Now we want to evaluate a double integral in general σ -finite measure, do we need to check the measurability and then define the double integral $f : \Omega_1 \times \Omega_2 \rightarrow \mathbb{R}$?
- Fubini's theorem gives the answer: No, we do not need to do this. Checking integrability property $\int_E |f(x)|\mu(dx) < \infty$ is enough.

Fubini-Tonelli's theorem

- **Theorem(Fubini's theorem).** Let $(\Omega_i, \mathcal{F}_i, P_i)$ for $i = 1, 2$ be σ -finite measure spaces and let $f : \Omega_1 \times \Omega_2 \rightarrow \mathbb{R}$ and $\int |f| dP_1 dP_2 < \infty$. For each $\omega_1 \in \Omega_1$ and $\omega_2 \in \Omega_2$, let $f_{\omega_1} : \Omega_2 \rightarrow \mathbb{R}$ and $f_{\omega_2} : \Omega_1 \rightarrow \mathbb{R}$ be defined by

- Main idea behind the proof.

Fubini-Tonellis's theorem

- Measure theory tells us: a nonnegative measurable function is always integrable. How to prove it? we need the upper and lower Lebesgue integrals definition, this is out of our scope.
- If we replace integrable of f by **non-negative measurable** f , the theorem is so-called Tonelli's theorem.
- Fubini's theorem tells if f is integrable, then

$$\int f dP = \int \left[\int f(\omega_1, \omega_2) dP_1(\omega_1) \right] dP_2(\omega_2) = \int \left[\int f(\omega_1, \omega_2) dP_2(\omega_2) \right] dP_1(\omega_1).$$

Tonelli's theorem tells that if $f \geq 0$ and is measurable, then the above holds.

- We also remark the Lebesgue integral and Riemann integral coincide on any bounded interval $[a, b]$ for the function f is Riemann integrable. So Fubini's and Tonelli's also work for Riemann integral for f is Riemann integrable.

Counter examples for Fubini's theorem

- **Definition(Counting measure).** All subsets of X are measurable sets. Then the counting measure μ on this measurable space (X, Σ) is the positive measure $\Sigma \rightarrow [0, +\infty]$ defined by

$$\mu(A) = |A|, \text{ if } A \text{ is finite,} \quad \mu(A) = +\infty, \text{ if } A \text{ is infinite,}$$

for all $A \in \Sigma$, where $|A|$ denotes the cardinality of the set A . The counting measure on (X, Σ) is σ -finite if and only if the space X is countable.

- If X is countable, e.g. $X = \mathbb{N}$, then the counting measure $c(A)$, which simply counts the number of points in a subset A . That is, $\mu(\{0, 1\}) = 2$, and $\mu(\{2, 4, 6, 8, \dots\}) = \infty$.
- **Example.** Let λ and μ be the lebesgue measure and counting measure, respectively on $([0, 1], \mathcal{B}([0, 1]))$. Then for $E = \{(x, y) \in \mathbb{R}^2 \mid y = x\}$

Counter examples for Fubini's theorem

- **Example.** Let $X = Y = \mathbb{N}$ and $\mu = \nu = \#$, where $\#$ is the counting measure. Then f is measurable, and f is integrable if and only if $\sum_n |f(n)| < \infty$ (and $\int_{\mathbb{N}} f d\# = \sum_n f(n)$). Now, consider a function $g : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ defined by

$$\begin{cases} g(n, n) = 1 & \forall n \geq 1 \\ g(n+1, n) = -1 & \forall n \geq 1 \\ g(m, n) = 0 & \text{for } m \neq n, m \neq n+1 \end{cases}$$

Hence, g is bounded and $\mu \times \nu$ -measurable.

An application of Fubini's theorem

- Fubini's theorem fails in evaluating the function $f(x, y) = \frac{y^2 - x^2}{(x^2 + y^2)^2}$ on $[0, 1] \times [0, 1]$.

- **Lemma(Moment of random variable).** In probability space $(\Omega, \mathcal{F}, \mathbb{P})$, if the random variable $Y \geq 0$ and the constant $p > 0$ then

$$\mathbb{E}(Y^p) = \int_0^\infty py^{p-1}P(Y > y)dy.$$

Independence of two events

- Probability space is $(\Omega, \mathcal{F}, \mathbb{P})$ where Ω is sample space, $\mathcal{F}$ is set of events (the σ -algebra) and $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is the probability function.
- Two events A and B are **independent** if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. Random variables X and Y are **independent** if for all sets C, D , we have $\mathbb{P}(X \in C, Y \in D) = \mathbb{P}(X \in C)\mathbb{P}(Y \in D)$, i.e., the events $\{X \in C\}$ and $\{Y \in D\}$ are independent. Two σ -fields $\mathcal{F}$ and $\mathcal{G}$ are independent if A and B are independent whenever $A \in \mathcal{F}$ and $B \in \mathcal{G}$.
- How to describe the independence over multiple n events?
- Let $(\Omega_i, \mathcal{F}_i), i \in \mathbb{N}$, be a sequence of measurable spaces. Let $(\prod_{\gamma \in \Gamma} \Omega_{\gamma}, \otimes_{\gamma \in \Gamma} \mathcal{F}_{\gamma})$ be its product, a measurable set $B \in \mathcal{F}_{\gamma_1} \otimes \mathcal{F}_{\gamma_2} \otimes \cdots \otimes \mathcal{F}_{\gamma_n}$, let the set B be chosen of the form $B = B_1 \times \cdots \times B_n$, for some $B_1 \in \mathcal{F}_1, \dots, B_n \in \mathcal{F}_n$, the set C is called a (product) **cylinder set**

Independence of multiple events

- For a product cylinder set C defined above and a fixed $n \in \mathbb{N}$, we specify the probability measure space $(\Omega_i, \mathcal{F}_i, P_i)$, $i \in \mathbb{N}$, we can define its probability by
- The equivalence between product measure and independence:
- Let Ω be the set of all countable sequences $\omega = (\omega_1, \omega_2, \omega_3 \dots)$ of real numbers. Let $\mathbb{P}$ be the probability measure that makes the ω_i independent identically distributed.
- Let $\mathcal{F}$ be the smallest σ -algebra (product σ -algebra) that makes the maps $\omega \rightarrow \omega_i$ measurable for $i = 1, 2, \dots, n$, the $\mathcal{F}$ is the product σ -algebra: smallest σ -algebra generated by the cylinder set of form $B_i = \{\omega_i \in (a_i, b_i], 1 \leq i \leq n\}$.
- Question: What we can say if the event that the ω_i converges to a limit? Let $n = \infty$, how the generated σ -algebra and the corresponding probability measure change?
- The limit exists, the marginal of the distribution is...?

- **Definition (Independence of Random Variables).** A collection of random variables $X_i : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (E_i, \mathcal{B}_i)$, $i \in I$ for some index set I , are independent if for any finite $J \subset I$ and any $B_j \in \mathcal{B}_j$,

$$\mathbb{P}(X_j \in B_j \text{ for } j \in J) = \prod_{j \in J} \mathbb{P}(X_j \in B_j).$$

- Pairwise independence: Any two events in the collection are independent of each other. Joint/mutual independence: Each event is independent of any combination of others.
- **Example.** The sample space Ω consisting of four equally likely points, a_1, a_2, a_3, a_4 . Let $A = \{a_1, a_2\}$, $B = \{a_2, a_3\}$, $C = \{a_3, a_1\}$. The three are not independent, but they are pairwise.

Example: multivariate normal distribution

- A random vector $(X_1, \dots, X_n)$ taking values in $\mathbb{R}^n$ is said to be a gaussian random vector if there exists $\mu = (\mu_1, \dots, \mu_n) \in \mathbb{R}^n$ and an $n \times n$ positive definite real symmetric matrix $\Sigma := (\sigma_{ij})_{1 \leq i, j \leq n}$, denoting $\det \Sigma$ to be determinant of Σ , it follows

$$\mathbb{P}((X_1, \dots, X_n) \in S) = \frac{1}{(2\pi)^{n/2}(\det \Sigma)^{1/2}} \int_S e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)} dx, \quad x \in S,$$

for all Borel sets $S \subset \mathbb{R}^n$ (where we identify elements of $\mathbb{R}^n$ with column vectors). The distribution of $(X_1, \dots, X_n)$ is called a **multivariate normal distribution**.

If $(X_1, \dots, X_n)$ is a gaussian random vector and $1 \leq i, j \leq n$:

- Bivariate case: show that X_i and X_j are pairwise independent if and only if the covariance $\text{Cov}(X_i, X_j)$ vanishes.
- Multivariate case: Furthermore, show that $(X_1, \dots, X_n)$ are jointly independent if and only if all the covariances $\text{Cov}(X_i, X_j)$ for $1 \leq i \neq j \leq n$ vanish.

Example: Bivariate case

Example: Multivariate case